

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
II Sem B.Tech. (CBCS) Mid Semester Examinations - I, February 2023

EC1001-1 – BASIC ELECTRONICS

Duration: 1 Hour

Max. Marks: 20

*Note: Answer any **One** full question from **each Unit**.*

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Explain the operation of a Zener voltage regulator with a neat circuit diagram. Also define the concepts of line and load regulation.	6	L*2	1	1
b)	In a full wave Bridge rectifier, the transformer secondary voltage is $100 \sin \omega t$ with 50 Hz supply frequency. The forward resistance of each diode is 25Ω and load resistance is 950Ω . Calculate (i) DC output voltage (ii) ripple factor and (iii) Efficiency. If now a capacitor of $100 \mu\text{F}$ is connected in parallel with the load to act as a filter, calculate the % ripple present at the output.	4	L3	1	1
2. a)	With neat circuit diagram and input/output waveforms, explain the operation of a Bridge rectifier. Derive the expressions for average value and r.m.s value of the load current.	6	L2	1	1
b)	For the diode circuit shown in Fig. 2. b), calculate the current I and the output voltage V_o .				

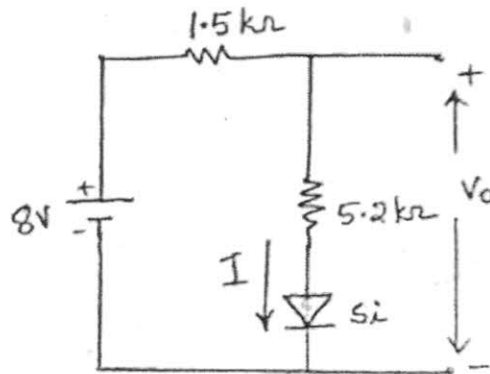


Fig. 2. b)

	Unit – II	Marks	BT*	CO*	PO*
3. a)	Explain the operation of an n-channel JFET with neat connection diagrams. Draw the drain and transfer characteristics for the same.	6	L2	2	1
b)	Determine α and β of a transistor for $I_c = 1 \text{ mA}$ and $I_B = 25 \mu\text{A}$. Considering the same value of β , calculate the new base current to generate $I_c = 5 \text{ mA}$.	4	L3	2	1
4. a)	With a neat circuit diagram of a CE-RC coupled amplifier, (i) Describe the significance of all the components (ii) Illustrate the phase reversal concept (iii) Draw the input/output waveforms	6	L2	2	1
b)	For a n-channel JFET, $I_{DSS} = 6 \text{ mA}$, $V_P = -4.5 \text{ V}$. Determine I_D at $V_{GS} = -2 \text{ V}$ and $V_{GS} = -6 \text{ V}$.	4	L3	2	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
